Few-Shot Hyperspectral Image Classification Using Relational Generative Adversarial Network

Wei Wei, Lei Tong, *Member, IEEE*, Baoqing Guo, Jun Zhou, *Senior Member, IEEE*, and Chuangbai Xiao

***Abstract*— Hyperspectral image (HSI) classification is an essential task in remote sensing, but its performance is greatly affected by limited labeled samples. Currently, generative adversarial network (GAN)-based methods can generate the virtue samples to augment the training set. However, with limited labeled data, GANs perform poorly in capturing features during sample generation. Very few relation networks (RNs) and few-shot learning (FSL) methods considered data augmentation to enhance performance. To address this challenge, we propose FSHyperRGAN, a few-shot HSI classification method based on relational GAN, which uses GANs to augment the training samples for RNs, while leveraging relation feature extraction to guide the generation of specific class samples. FSHyperRGAN comprises four modules: a data processing module converting the HSI data to 1-D and 3-D features, an adversarial generation (AG) module synthesizing virtual samples conditioned on labels, a data embedding and reconstruction (DER) module encoding latent spaces for accurate sample reconstruction while preserving category characteristics, and a relation computation (RC) module computing relation scores across generated, reconstructed, and original samples. In addition, a relational feature matching scheme is also applied, which can use virtual samples to guide classification. Two FSHyperRGAN frameworks are designed, 1D-FSHyperRGAN and 3D-FSHyperRGAN, which can be utilized for 1-D spectral or 3-D spatial–spectral classification. Experiments on widely used HSI datasets illustrate that the proposed method outperforms several state-of-the-art methods and achieves overall accuracies of 92.64%, 86.82%, 83.64%, and 84.57% on KSC, PaviaU, Houston, and WHU-Hi-HongHu datasets, respectively.**

***Index Terms*— Deep learning, few-shot learning (FSL), generative adversarial network (GAN), hyperspectral image (HSI) classification.**

I. Introduction

HYPERSPECTRAL imaging is an important remote sensing technique that captures information on objects in hundreds of continuous spectral bands. The rich spectral information of hyperspectral images (HSIs) can improve the ability to distinguish and identify features, a wide range of applications such as agriculture, disease diagnosis, mineral exploration, urban planning, and disaster management [1], [2], [3], [4], [5].

HSI classification (HIC) utilizes both spectral and spatial information to determine the category of each pixel. Over the past few decades, various methods have been proposed for HIC using spectral information in HSI, including machine learning methods such as support vector machine (SVM) [6], [7], [8], k-nearest-neighbor classifier [9], dynamic or random subspace [10], [11], neural network [12], and logistic regression [13], [14]. With the improvement of spatial resolution in hyperspectral data, rich spatial information can significantly improve the classification performance [15]. Multiple kernel learning (MKL) has been successfully applied to HSI classification, capable of handling the heterogeneous fusion of spectral and spatial features [16]. However, for these traditional methods, manually extracted features involve massive parameter settings, which affect the classification accuracy.

In recent years, deep learning methods have been widely applied to HIC, with two types of feature extraction methods. One type of method extracts spatial and spectral features through two branches. Yang et al. [17] extracted the features from spectral and spatial domains and fused them into fully connected layers for classification. Gao et al. [18] proposed a spectral–spatial residual network (SDRN) which stacked spectral and spatial branches to improve the efficiency and accuracy. Li et al. [19] proposed two 2-D CNN networks to simultaneously extract spectral, local spatial features, and global spatial features to enhance the classification performance. Another type of method simultaneously extracts spatial–spectral features. Hamida et al. [20] proposed a 3D-CNN for simultaneous spatial–spectral feature extraction without preprocessing. Zhu et al. [21] introduced a deep convolutional capsule network (Conv-Capsule) designed for HSI classification, better extracting hierarchical relationships, and spatial structures. Xie et al. [22] proposed a multiscale approach to better capture spatial features. In addition to the aforementioned models based on various convolutional neural networks, some studies have employed automatic neural architecture search algorithms [23], [24], [25] to automatically design neural network architectures that better adapt to HIC. Graph convolutional neural networks [26], [27] have also been applied to HIC, leveraging graph structures to capture the relationships and similarities among hyperspectral data. Cai et al. [28] proposed a deep two-step low-shot learning

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framework to accurately classify using limited training images. Sun et al. [29] proposed a large kernel spectral spatial attention network (LKSSAN) to capture the long-range 3-D characteristics of HSI, making full use of the dependency between spectral and spatial information. Although HIC methods based on deep learning have achieved high classification accuracy, they still face the problem of limited training data, which severely affect the classification accuracy.

In order to solve this problem, researchers have proposed GAN-based methods [30] to generate virtual samples to augment the training data [31], [32], [33], [34]. The pioneering GAN-based HSI classification method proposed by Zhu et al. [35] demonstrated practicality and effectiveness. Feng et al. [36] proposed a multiclass spatial–spectral feature GAN with separate spectral and spatial generators, feeding virtual features into the discriminator for classification. A triplet generative adversarial network (GAN) was used for 1-D structure generation, with capsule network adopted for classification [37]. To optimize GAN training, Zhang et al. [32] combined PG-GAN [38] and Wasserstein GAN with gradient penalty (GP) [39]. Chen et al. [40] innovatively integrated self-attention with conditional variational autoencoder (VAE) GAN, achieving improved generation results. Lv et al. [41] proposed a superpixel-constrained sample enhancement method, which was named ISGBA by iteratively generating and balancing initial training samples and combining them with pattern recognition neural networks for classification. However, with limited labeled data, although GANs can generate virtual samples, they perform poorly in capturing features during sample generation.

Another way to solve limited training samples problem is using few-shot learning (FSL) [42], [43], [44], [45], [46]. FSL aims to enable models to generalize to new tasks with only a few training examples. Due to its outstanding generalization capability, this strategy has been extensively applied to FSL tasks such as HIC [47], [48], [49], [50], [51]. Liu et al. [47] proposed a deep FSL (DFSL), through deep residual 3-D CNN to extract spatial–spectral features, and finally used SVM for classification. Deng et al. [48] proposed an FSL method based on metric learning. The embedded features are obtained through deep convolution of the samples, and the similarity between samples was learned according to the deep embedded features to classify unlabeled data. Li et al. [49] enhanced few-shot discriminative learning by combining channel attention and spatial attention modules with feature networks using different feature fusion schemes. Zuo et al. [50] made the first attempt to explicitly quantify the correlation between pixels using edge label graph neural network in HSI FSL, modeling the intraclass similarity and interclass dissimilarity between HSI pixels. Deep cross-domain FSL (DCFSL) [51] discovered transferable knowledge in the source class and learned the discriminative embedding model of the target class [52], which showed better classification results than DFSL. Zeng et al. [53] proposed a novel dual-metric multistage relation network (DM-MRN) for small-sample HSI classification, integrating multiple relation scores. However, these methods ignored the data augmentation methods to enhance the performance.

To address the above two issues, we propose to combine the data augmentation methods and the FSL method to enhance the classification accuracy with few training data. In this article, we propose FSHyperRGAN, shown in Fig. 1, which employs a generative model and relation network (RN) to enhance the classification accuracy for few-shot HIC. Generative model can generate virtual HSI samples to augment the training set, while RN can better extract features. Moreover, RN computes the similarity between samples, which can obtain more robust classification accuracy.

The main contributions of the article are summarized as follows.

1) We propose a novel method FSHyperRGAN, which integrates the generative model and the relational network for few-shot HIC. Two HIC frameworks are proposed: a 1-D spectral classification network and a 3-D spatial–spectral classification network, which can be used for 1-D spectral and 3-D spectral–spatial HIC.
2) In the generation process, we design a relational feature matching scheme, which can generate virtual samples with stronger specific class features and use virtual samples to guide classification.
3) In the relational computation process, the relational computation module enhances classification accuracy by integrating a class feature extractor and relation calculator, augmenting data with generated and reconstructed samples, and combining losses to optimize relational computations.

The remainder of this article is organized as follows. The related work is introduced briefly in Section II. The FSHyperRGAN is described in detail in Section III. The experimental results and analysis are given in Section IV. Finally, the conclusion is drawn in Section V.

II. Related Work

In this section, we review VAE/GAN and RN, which form the foundation for our proposed FSHyperRGAN.

A. VAE/GAN

As shown in Fig. 2, VAE/GAN [54] combines VAE and GAN, using the GAN discriminator to guide the VAE to reconstruct the target. It contains three parts: 1) an encoder network $\boldsymbol{E}$; 2) a generative network $\boldsymbol{G}$; and 3) a discriminative model $\boldsymbol{D}$.

The VAE module in VAE/GAN consists of two networks. The encoder $\boldsymbol{E}$ encodes the input data $\mathbf{x}$ into a potential representation $\mathbf{z}$, and the generator $\boldsymbol{G}$ decodes the potential representation to obtain the data space $\tilde{\mathbf{x}}$. Assuming that the input data $\mathbf{x}$ has a distribution $p(\mathbf{x})$, the goal of the generator $\boldsymbol{G}$ in the VAE module is to make the distribution $p(\mathbf{x}|\mathbf{z})$ of the generated data $\tilde{\mathbf{x}}$ to fit $p(\mathbf{x})$ as much as possible, that is, the purpose of the VAE is to rebuild $\mathbf{x}$. Therefore, the distribution of $\mathbf{z}$ and $\tilde{\mathbf{x}}$ can be expressed as follows:

$$\mathbf{z} \sim \boldsymbol{E}(\mathbf{x}) = q(\mathbf{z}|\mathbf{x}), \quad \tilde{\mathbf{x}} \sim \boldsymbol{G}(\mathbf{z}) = p(\mathbf{x}|\mathbf{z}). \tag{1}$$

B. Relation Network

In order for the generator $\boldsymbol{G}$ to rebuild $\mathbf{x}$ from $\mathbf{z}$ sampled from the distribution $q(\mathbf{z}|\mathbf{x})$, the VAE regularizes the encoder

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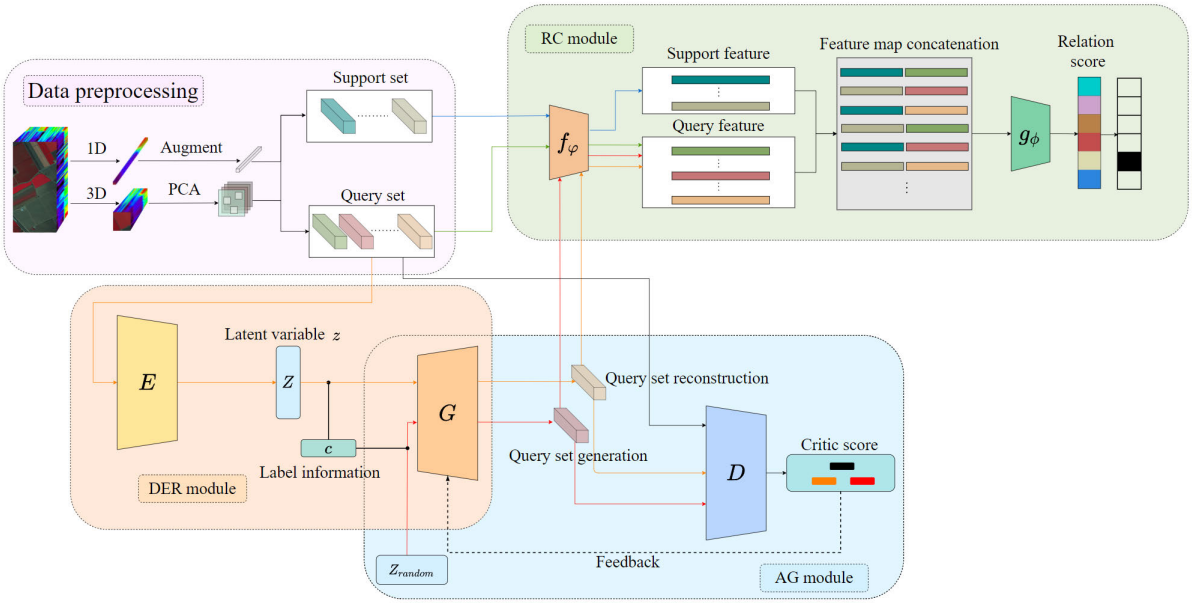


Fig. 1. Illustration of the overall structure of the FSHyperRGAN framework. The FSHyperRGAN consists of four main modules: the data preprocessing module, the AG module, the DER module, and the RC module.

by imposing a prior on the latent distribution $p(\mathbf{z})$. Usually, normal distribution $\mathbf{z} \sim \mathcal{N}(\mu, \sigma^2)$ is chosen. The VAE loss is the sum of the expected log likelihood (the reconstruction error) and a prior regularization term

$$\mathcal{L}_{\text{VAE}} = -\mathbb{E}_{q(\mathbf{z}|\mathbf{x})}\left[\log \frac{p(\mathbf{x} \mid \mathbf{z})p(\mathbf{z})}{q(\mathbf{z} \mid \mathbf{x})}\right] = \mathcal{L}_{\text{llike}}^{\text{pixel}} + \mathcal{L}_{\text{prior}} \quad (2)$$

with

$$\mathcal{L}_{\text{llike}}^{\text{pixel}} = -\mathbb{E}_{q(\mathbf{z}|\mathbf{x})}[\log p(\mathbf{x} \mid \mathbf{z})] \quad (3)$$

$$\mathcal{L}_{\text{prior}} = D_{\text{KL}}(q(\mathbf{z} \mid \mathbf{x}) \| p(\mathbf{z})) \quad (4)$$

where D_{KL} is the Kullback–Leibler (KL) divergence.

In the GAN network, the generation network $\boldsymbol{G}(\mathbf{z})$ maps the latent variable $\mathbf{z}$ to the data space, and the output $\boldsymbol{D}(\mathbf{x})$ of the discriminator $\boldsymbol{D}$ represents the probability that the input $\mathbf{x}$ is a real picture. The goal of the generation network $\boldsymbol{G}$ is to generate real pictures as much as possible to deceive the discriminant network $\boldsymbol{D}$, and the goal of the binary classifier $\boldsymbol{D}$ is to separate the pictures generated by $\boldsymbol{G}$ from the real pictures. The realization method of GAN is to train the generator model and the discriminator model against each other, and finally reach the Nash equilibrium. Therefore, GAN aims to maximize/minimize the binary cross entropy

$$\min_{\boldsymbol{G}} \max_{\boldsymbol{D}} \mathcal{L}_{\text{GAN}} = \mathbb{E}_{\mathbf{x}\sim p(\mathbf{x})}\big[\log \boldsymbol{D}(\mathbf{x})\big] + \mathbb{E}_{\mathbf{z}\sim p(z)}\big[\log 1 - \boldsymbol{D}(\boldsymbol{G}(\mathbf{z}))\big]. \quad (5)$$

After adding the GAN model to VAE, the discriminator network can obtain rich hidden features of the image. To achieve this, $\boldsymbol{D}_l(\boldsymbol{x})$ denotes the hidden representation of the lth layer of the discriminator. VAE reconstruction error of (3) is replaced by

$$\mathcal{L}_{\text{llike}}^{\boldsymbol{D}_l} = -\mathbb{E}_{q(\mathbf{z}|\mathbf{x})}\big[\log p(\boldsymbol{D}_l(\mathbf{x}) \mid \mathbf{z})\big]. \quad (6)$$

Finally, the VAE/GAN is trained by the triple criterion loss function

$$\mathcal{L}_{\text{VAE/GAN}} = \mathcal{L}_{\text{prior}} + \mathcal{L}_{\text{llike}}^{\boldsymbol{D}_l} + \min_{\boldsymbol{G}} \max_{\boldsymbol{D}} \mathcal{L}_{\text{GAN}}. \quad (7)$$

RN [55] belongs to metric learning methods in FSL, shown in Fig. 3, with the objective of mapping the input space to a new embedding space. In this embedding space, a similarity measure is employed to distinguish between different classes. In FSL, the model is usually trained by the episodes $\mathbf{T}$ proposed in [42]. In each training iteration, each episode $\mathbf{T_i}$ samples a task $\mathbf{\Gamma}_i$, a part of data from N classes randomly selected from the training set. Each task $\mathbf{\Gamma}_i$ contains a *support set* $\mathbf{S}$, and a *query set* $\mathbf{Q}$. The *support set* $\mathbf{S} = \{(\mathbf{x}_i, \mathbf{y}_i)\}_{i=1}^{m}$ is composed of K labeled samples from each of the N classes, while the remaining part of each class serves as the *query set* $\mathbf{Q} = \{(\mathbf{x}_j, \mathbf{y}_j)\}_{j=1}^{n}$. The test set follows a similar setup. The core idea of RN is to complete classification by computing similarity scores between *query set* samples and *support set* samples in an N-way, K-shot setting. RN is composed of embedding module $\boldsymbol{f}$ and a relation module g. The samples $\mathbf{x}_i$ from the *query set* and $\mathbf{x}_j$ from the *support set* are passed through the embedding module to generate feature mappings $\boldsymbol{f_\varphi}(\mathbf{x}_i)$ and $\boldsymbol{f_\varphi}(\mathbf{x}_j)$, respectively. For K-shot where K>1, the embedding module outputs of all samples in the *support set* for each training class are summed element-wise to form feature maps for each class. The feature maps $\boldsymbol{f_\varphi}(\mathbf{x}_i)$ and $\boldsymbol{f_\varphi}(\mathbf{x}_j)$ obtain the concatenation of feature maps by operation C. Ultimately, the feature map of the combination of *query set* samples and *support set* samples is fed into the relation module $\boldsymbol{g}$ to obtain relation scores $\mathbf{r}_{i,j} \in [0, 1]$, which indicates the similarity between $\mathbf{x}_i$ and $\mathbf{x}_j$

$$\mathbf{r}_{i,j} = \boldsymbol{g_\phi}(\mathcal{C}(\boldsymbol{f_\varphi}(\mathbf{x}_i), \boldsymbol{f_\varphi}(\mathbf{x}_j)), \quad i = 1, 2, 3, \ldots, N. \quad (8)$$

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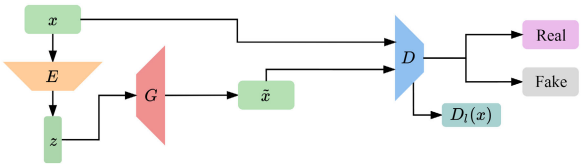


Fig. 2. Architecture of the VAE/GAN model.

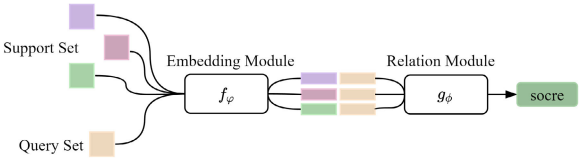


Fig. 3. Architecture of the RN model.

The RN ultimately treats classification as a regression task and trains the model using mean squared error (MSE)

$$\varphi, \phi \leftarrow \text{argmin}_{\varphi,\phi} \sum_{i=1}^{I} \sum_{j=1}^{J} (\mathbf{r}_{i,j} - \mathbf{1}(\mathbf{y}_i == \mathbf{y}_j))^2. \quad (9)$$

III. Methodology

In this section, we describe our proposed FSHyperRGAN in detail. From Fig. 1, we can see that the proposed FSHyperRGAN is mainly composed of four modules: data preprocessing, adversarial generation (AG), data embedding and reconstruction (DER), and relation computation (RC). In the data preprocessing module, it denoises and removes outliers from HSI data and splits the data into support and query sets for FSL. In the AG module, the generator $\boldsymbol{G}$ synthesizes virtual samples conditioned on labels, which the discriminator $\boldsymbol{D}$ learns to distinguish from real samples. In the DER module, a conditional VAE learns the latent feature space for accurate sample reconstruction, guided by a class feature extractor $\boldsymbol{f_\varphi}$ for preserving category characteristics. Finally, in the RC module, relation scores are computed across the generated virtual samples, reconstructed samples, and original samples, leveraging both generative and reconstructive features to enhance the final RN $\boldsymbol{g_\phi}$. The computed relation scores, in turn, provide feedback to optimize the generation and reconstruction processes.

A. Data Preprocessing

To better extract the spectral and spectral–spatial data, the data preprocessing module converts the hyperspectral data to 1-D and 3-D domains. For the 1-D data, to fully utilize the spectral information, we retain all band information from the original HSI. However, due to noise and outliers in HSI, the module first removed outliers and then normalized the spectral data by bands. To augment the spectral data, we also mix the spectral as follows:

$$\mathbf{x}_{\text{new}} = \lambda \cdot \mathbf{x}_1 + (1 - \lambda) \cdot \mathbf{x}_2, \quad \lambda \sim \text{Be}(0.1, 0.1) \quad (10)$$

where Be means beta distribution, x_1 and x_2 are different spectral samples from the same class, and x_{new} is the data merged by x_1 and x_2 to enhance the model generation ability.

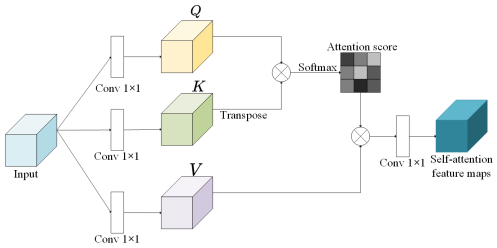


Fig. 4. Illustrations of the attention layers utilized in FSHyperRGAN.

For the 3-D data, the principal component analysis (PCA) is used to reduce the dimensionality of the HSI data. This not only reduces the huge phenomenon but also makes it easy to generate the synthetic data.

B. AG Module

The AG module consists of a generator $\boldsymbol{G}$ and a discriminator $\boldsymbol{D}$. The generator $\boldsymbol{G}$ is used to generate simulated HSIs, while the discriminator $\boldsymbol{D}$ distinguishes between real and generated HSIs. Depending on the data, two different types of $\boldsymbol{G}$ and $\boldsymbol{D}$ were designed: 1D-$\boldsymbol{G}$, 3D-$\boldsymbol{G}$, 1D-$\boldsymbol{D}$, and 3D-$\boldsymbol{D}$. The 1D-$\boldsymbol{G}$ network is composed of four convolutional layers, while the 3D-$\boldsymbol{G}$ consists of seven layers, including one fully connected layer, two self-attention layers, and deconvolution layers. The self-attention layer shown in Fig. 4 captures the weights of the sample space dimensions by calculating attention scores, thereby generating self-attention feature maps. In pixel-level classification, this mechanism not only focuses on local spatial features around the central pixel but also captures long-range spatial dependencies based on sample conditions, such as class boundaries or spatial intersections of pixels from different classes. This enables our model to flexibly learn and utilize spatial information. The 1D-$\boldsymbol{D}$ network is composed of five convolutional layers and one fully connected layer, while the 3D-$\boldsymbol{D}$ consists of seven layers, including five convolutional layers and one fully connected layer. Compared to the 1-D structure, the 3-D structure uses 3×3 convolutional kernels to extract spatial features. Their network structures are shown in Fig. 5.

To adapt to the few-shot setting and generate better simulated data, for each training episode $\mathbf{T_i}$, we sample a task $\boldsymbol{\Gamma}_i$ from the training set and use the label information $\mathbf{c}$ in the query set $\mathbf{Q}$ to guide the generation of samples from the random latent variable $\mathbf{z_{rand}}$, which comprises random numbers from a standard normal distribution, serving as a prior to regulate the encoder 3D-$\boldsymbol{E}$. The label information $\mathbf{c}$ is the one-hot encoding of the label $\mathbf{y}_j$ of the original sample $\mathbf{x}_j$.

To further simplify the computation and accelerate the training process, we use the Wasserstein distance to train the discriminator $\boldsymbol{D}$, as shown in the following:

$$\mathcal{L}_{\boldsymbol{D}} = \mathbb{E}_{\mathbf{x}_j \sim p(\mathbf{x})}\left[\boldsymbol{D}\left(\mathbf{x}_j\right)\right] - \mathbb{E}_{\mathbf{x}_j^{\text{gen}} \sim p(\tilde{\mathbf{x}})}\left[\boldsymbol{D}\left(\mathbf{x}_j^{\text{gen}}\right)\right] \quad (11)$$

where $\mathbf{x}_j$ is the original data sample, and $\mathbf{x}_j^{\text{gen}}$ is the synthetic sample generated by the network $\boldsymbol{G}$: $\mathbf{x}_j^{\text{gen}} = \boldsymbol{G}(\mathbf{z_{rand}}, \mathbf{c})$. The generated sample $\mathbf{x}_j^{\text{gen}}$, along with its original label $\mathbf{y}_j$, form the query set generated data $\mathbf{Q}^{\text{gen}}$. Adding label information

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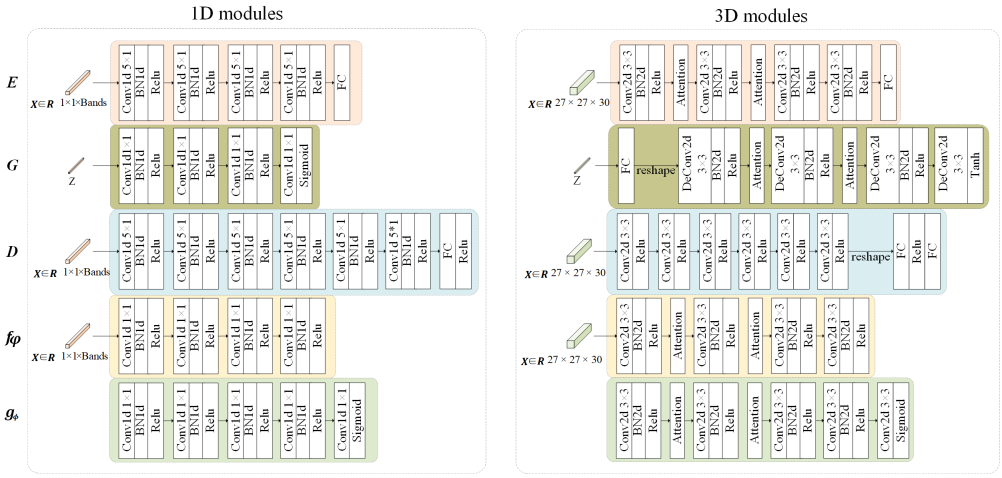


Fig. 5. Detailed 1-D and 3-D design illustrations of the various modules in FSHyperRGAN.

only affects the input dimension of $\boldsymbol{G}$, but does not affect the dimension of generated and reconstructed data. The discriminator $\boldsymbol{D}$'s ability to distinguish between real and generated data is measured by the distribution distance between $\mathbf{x}_j$ and $\mathbf{x}_j^{\text{gen}}$, reducing the computational complexity of logarithms and other operations compared to the original GAN formula.

To further stabilize the training and generate higher quality images, we use GP to train the network. By penalizing cases where the gradient norm is greater than 1, we can enhance the smoothness of the discriminator in the high-dimensional space and avoid abrupt gradient changes. Moreover, to ensure that the generator network $\boldsymbol{G}$ has a well-trained discriminator $\boldsymbol{D}$, we train the discriminator $\boldsymbol{D}$ multiple times while training the generator $\boldsymbol{G}$ at a lower frequency. After $\mathbf{d_{step}}$ iterations of training the discriminator, the generator is trained, and GP is applied. The loss function of the AG module can be expressed as follows:

$$\mathcal{L}_{AG} = L_{\boldsymbol{D}} + \underbrace{\lambda\left[\left(\left\|\nabla_{\mathbf{x}_j^{\text{gen}}} D\left(\mathbf{x}_j^{\text{gen}}\right)\right\|_2 - 1\right)^2\right]}_{\text{Gradient Penalty}} \tag{12}$$

where $\nabla_{\mathbf{x}_j^{\text{gen}}}$ represents the gradient of the sample $\mathbf{x}_j^{\text{gen}}$ generated by $\boldsymbol{G}$, and λ controls the strength of the GP.

C. DER Module

The DER module consists of an encoder $\boldsymbol{E}$ and a generator $\boldsymbol{G}$. The encoder $\boldsymbol{E}$ maps the hyperspectral data to a low-dimensional feature space, while $\boldsymbol{G}$ reconstructs the HSI. Additionally, we use a feature extractor $\boldsymbol{f_\varphi}$ to extract HSI features. Depending on the data, different $\boldsymbol{E}$ and $\boldsymbol{f_\varphi}$ were designed, namely 1D-$\boldsymbol{E}$, 3D-$\boldsymbol{E}$, 1D-$\boldsymbol{f_\varphi}$, and 3D-$\boldsymbol{f_\varphi}$. The 1D-$\boldsymbol{E}$ network is composed of four convolutional layers and one fully connected layer, while the 3D-$\boldsymbol{E}$ consists of seven layers, including two self-attention layers, four convolutional layers, and one fully connected layer. The self-attention layers are used here to compensate for the lack of long-range spatial dependencies in convolutional networks. The 1D-$\boldsymbol{f_\varphi}$ and 3D-$\boldsymbol{f_\varphi}$ networks are each composed of four convolutional layers. Their network structures are shown in Fig. 5.

The encoder $\boldsymbol{E}$ learns the latent features $\mathbf{z} = \boldsymbol{E}(\mathbf{x}_j)$ from the data $\mathbf{x}_j$ in the query set $\mathbf{Q}$. The generator $\boldsymbol{G}$ decodes $\mathbf{z}$ together with the label information $\mathbf{c}$ to generate the reconstructed sample $\mathbf{x}_j^{\text{recon}} = \boldsymbol{G}(\mathbf{z}, \boldsymbol{c})$. To constrain the encoder to produce meaningful latent representations $\mathbf{z}$ and obtain better reconstruction quality, we use the KL divergence to constrain the distribution of $\mathbf{z}$ as follows:

$$\mathcal{L}_z = -\frac{1}{2}\sum_{j=1}^{J}(1 + \log(\sigma^2) - \mu^2 - \sigma^2) \tag{13}$$

where the encoder $\boldsymbol{E}$ encodes $\mathbf{x}_j$ into the mean μ and standard deviation σ, which parameterize the posterior distribution of the latent variable z, i.e., $\mathbf{z}$ is sampled from a normal distribution with μ and σ, constraining the distance between the latent variable distribution $\mathbf{z}$ and the predefined prior distribution (assume $\mathbf{x}_j$ follows a standard Gaussian distribution).

To measure the distance between the original sample $\mathbf{x}_j$ and the reconstructed sample $\mathbf{x}_j^{\text{recon}}$, we adopt the MSE loss to represent the difference between them. The loss function of the reconstructed sample can be defined as follows:

$$\mathcal{L}_{\text{recon}} = \mathcal{L}_z + \frac{1}{J}\sum_{j=1}^{J}\left(x_j - x_j^{\text{recon}}\right)^2. \tag{14}$$

To ensure the diversity of the samples generated by $\boldsymbol{G}$, we further train $\boldsymbol{D}$ on the generated samples from $\boldsymbol{G}$. To make the simulated samples better assist the classification task, we use the class feature extractor $\boldsymbol{f_\varphi}$ from the relational module to enhance the generated network's attention to class differential information. For this purpose, we incorporate the class feature loss into the data reconstruction, capturing the difference between $\boldsymbol{f_\varphi}(\mathbf{x}_j)$ and $\boldsymbol{f_\varphi}(\mathbf{x}_j^{\text{recon}})$. Additionally, we perform class determination on the features extracted by $\boldsymbol{f_\varphi}$ within the RC module. The final loss function of the DER module is defined as follows:

$$\mathcal{L}_{\text{DER}} = \mathcal{L}_{\text{recon}} - \mathbb{E}_{\mathbf{x}_j^{\text{recon}}\sim p(\tilde{\mathbf{x}})}\left[\boldsymbol{D}\left(\mathbf{x}_j^{\text{recon}}\right)\right] + \frac{1}{2}\left\|\boldsymbol{f_\varphi}(\mathbf{x}_j) - \boldsymbol{f_\varphi}\left(\mathbf{x}_j^{\text{recon}}\right)\right\|_2^2 \tag{15}$$

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where $\boldsymbol{D}(\mathbf{x}_j^{\text{recon}})$ represents the further training of $\boldsymbol{D}$ on the reconstructed samples. By training the discriminator $\boldsymbol{D}$, we can encourage the generator $\boldsymbol{G}$ to better utilize the entire latent space, rather than staying in a local area.

D. RC Module

The RC module consists of a class feature extractor $\boldsymbol{f_\varphi}$ and a relation calculator $\boldsymbol{g_\phi}$. The $\boldsymbol{f_\varphi}$ extracts class features, while $\boldsymbol{g_\phi}$ performs relational computation between query samples and support samples. Depending on the data, we designed different $\boldsymbol{g_\phi}$. Both the 1D-$\boldsymbol{g_\phi}$ and 3D-$\boldsymbol{g_\phi}$ networks are composed of five convolutional layers. The 1D-$\boldsymbol{g_\phi}$ uses 1×1 convolutional kernels to extract spectral features, while the 3D-$\boldsymbol{g_\phi}$ uses 3×3 convolutional kernels to additionally extract spatial features. Their network structures are shown in Fig. 5.

To enhance the accuracy of the calculation module and improve the classification effect when samples are scarce, we augmented the data with the generated samples from the previous two modules, obtaining the virtually generated samples $\mathbf{Q}^{\text{gen}} = \{(\mathbf{x}_j^{\text{gen}}, \mathbf{y}_j)\}_{j=1}^n$ and the reconstructed samples $\mathbf{Q}^{\text{recon}} = \{(\mathbf{x}_j^{\text{recon}}, \mathbf{y}_j)\}_{j=1}^n$. We then concatenate the feature vectors of the support set with the feature vectors of the query set denoted as $\mathcal{C}$ after dimension expansion, and feed these concatenated feature vectors into $\boldsymbol{g_\phi}$ for computation to obtain the relational scores for the query set data as follows:

$$\mathbf{r}_{i,j} = \boldsymbol{g_\phi}\big(\mathcal{C}\big(\boldsymbol{f_\varphi}(\mathbf{x}_i), \boldsymbol{f_\varphi}\big(\mathbf{x}_j\big)\big)\big) \tag{16}$$

$$\mathbf{r}_{i,j}^{\text{recon}} = \boldsymbol{g_\phi}\big(\mathcal{C}\big(\boldsymbol{f_\varphi}(\mathbf{x}_i), \boldsymbol{f_\varphi}\big(\mathbf{x}_j^{\text{recon}}\big)\big)\big) \tag{17}$$

$$\mathbf{r}_{i,j}^{\text{gen}} = \boldsymbol{g_\phi}\Big(\mathcal{C}\Big(\boldsymbol{f_\varphi}(\mathbf{x}_i), \boldsymbol{f_\varphi}\Big(\mathbf{x}_j^{\text{gen}}\Big)\Big)\Big) \tag{18}$$

After obtaining the relation scores for the original sample query set, the synthetic sample query set, and the reconstructed sample query set, in order for the network $\boldsymbol{g_\phi}$ to serve as an effective metric function, we combine the losses calculated from the classification results of the samples in each query set, as shown below

$$\mathcal{L}_{\text{RC}} = \sum_{i=1}^{I}\sum_{j=1}^{J}(\mathbf{r}_{i,j} + \alpha \cdot \mathbf{r}_{i,j}^{\text{recon}} + \beta \cdot \mathbf{r}_{i,j}^{\text{gen}} - \mathbf{3}(\mathbf{y}_i == \mathbf{y}_j))^2 \tag{19}$$

where the parameters α and β control the respective contributions of the reconstructed and generated samples to the relational computation for classification. In the earlier rounds, the quality of the generated samples is relatively poor, so we gradually increase α and β as the number of training rounds increases, enhancing the relational computation's dependence on the generated samples. The overall process of the proposed method FSHyperRGAN is shown in Algorithm 1.

Algorithm 1 Training Pipeline of the Proposed FSHyperRGAN Algorithm

Require: Input Hyperspectral Image;
Ensure: Divided into training data and test data;
Generated virtual sample;
Predicted label for each pixel;

for each episode $\mathbf{T_i}$ **do**
  Sample $\mathbf{\Gamma}_{\text{i}}$ in episode $\mathbf{T_i}$ from the training data;
  for $i = 1$ to $\mathbf{d_{step}}$ **do**
    Obtain the $\mathbf{Q}^{\text{gen}}$ data using $\boldsymbol{G}$ based on the random variable $\mathbf{z_{rand}}$;
    Using $\mathbf{Q}$ and $\mathbf{Q}^{\text{gen}}$ for adversarial generation training according to Eq.(12) to calculate the loss function $\mathcal{L}_{AG_{stage}}$;
    $\theta_{\boldsymbol{D}} \overset{+}{\leftarrow} -\nabla_{\theta_{\boldsymbol{D}}}(\mathcal{L}_{AG})$
  end for
  Obtain the reconstructed $\mathbf{Q}^{\text{recon}}$ data of $\mathbf{Q}$ through the encoder $\boldsymbol{E}$ and decoder G;
  Calculate the reconstruction loss $\mathcal{L}_{DER_{stage}}$ according to Eq.(15);
  $\theta_{\boldsymbol{E}}, \theta_{\boldsymbol{G}}, \theta_{\boldsymbol{D}}, \varphi \overset{+}{\leftarrow} -\nabla_{\theta_{\boldsymbol{E},\boldsymbol{G},\boldsymbol{D},\varphi}}(\mathcal{L}_{DER})$
  Obtain the relation scores $\mathbf{r}$, $\mathbf{r}^{\text{recon}}$, $\mathbf{r}^{\text{gen}}$ between $\mathbf{Q}$, $\mathbf{Q}^{\text{recon}}$, $\mathbf{Q}^{\text{gen}}$ and S through $\boldsymbol{f_\varphi}$ and $\boldsymbol{g_\phi}$;
  Calculate the relation loss $\mathcal{L}_{RC_{stage}}$ according to Eq.(19);
  $\varphi, \phi, \theta_E, \theta_G \overset{+}{\leftarrow} -\nabla_{\theta_{\varphi,\phi,E,G}}(\mathcal{L}_{RC})$
end for

IV. Experiments

In this section, we conduct the experiments with our FSHyperRGAN on four public HSI datasets, including the Kennedy Space Center (KSC) dataset, the Pavia University (PaviaU) dataset, the Houston dataset, and the WHU-Hi-HongHu dataset. The classification performance of the experiment is assessed by overall accuracy (OA), average accuracy (AA), and Kappa coefficients.

TABLE I
Land Cover Classes and the Numbers of Samples in the KSC Dataset

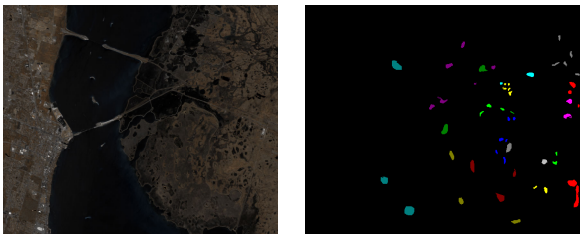
Class No.	Color	Class Name	Pixels
1		Scrub	761
2		Willow swamp	243
3		Cabbage palm hammock	256
4		Cabbage palm/oak hammock	252
5		Slash pine	161
6		Oak/broadleaf hammock	229
7		Hardwood swamp	105
8		Graminoid marsh	431
9		Spartine marsh	520
10		Cattail marsh	404
11		Salt marsh	419
12		Mud flats	503
13		Water	927
		Total	5211

A. HSI Datasets Description

1) Kennedy Space Center: The dataset was collected by the NASA Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) instrument over the KSC, FL, USA. The raw spatial dimensions of the data are 512×614, with a relatively low spatial resolution of 18 m. It encompasses 13 classes representing different pixel land cover materials. The dataset contains 5211 labeled data points, as shown in Table I. Fig. 6 displays the false color image and ground-truth map of the KSC dataset.

2) Pavia University: The PaviaU dataset is a high-resolution HSI dataset captured in Pavia, Italy, using a German

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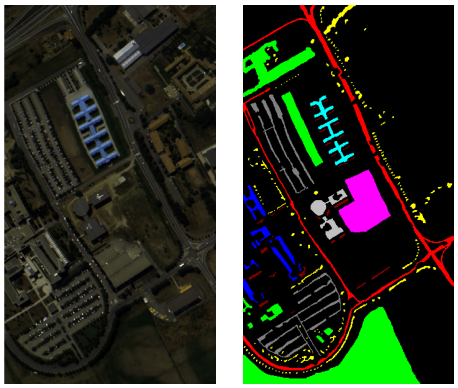


(a) (b)

Fig. 6. KSC dataset. (a) False color image. (b) Ground-truth map.

TABLE II
LAND COVER CLASSES AND THE NUMBERS OF SAMPLES IN THE PAVIAU DATASET

Class No.	Color	Class Name	Pixels
1		Asphalt	6631
2		Meadows	18649
3		Gravel	2099
4		Trees	3064
5		Painted metal sheets	1345
6		Bare Soil	5029
7		Bitumen	1330
8		Self-Blocking Bricks	3682
9		Shadows	947
		Total	42506



(a) (b)

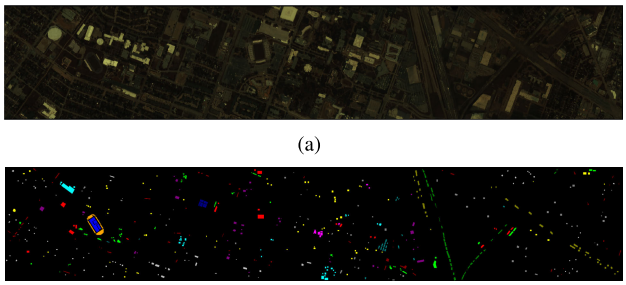
Fig. 7. PaviaU dataset. (a) False color image. (b) Ground-truth map.

airborne reflective optical spectroscopy imaging instrument in 2003. The PaviaU dataset has a size of 610 × 340 pixels and consists of nine major classes with a total of 207 400 pixels. However, the background pixels constitute a significant portion, with only 42 766 pixels representing ground objects. The land cover categories and sample quantities for the PaviaU dataset are presented in Table II. The false color image and ground-truth map of PaviaU are shown in Fig. 7.

3) Houston: The Houston dataset was provided by the HSI Analysis Group and the NSF-funded Center for Airborne Laser Mapping (NCALM) at the University of Houston. The dataset has a size of 349 × 1905 and includes 144 spectral bands covering a wavelength range from 364 to 1046 nm. The land cover vegetation is annotated into 15 classes. Table III shows the land cover classes and samples' size of the Houston. The false color image and ground-truth map of Houston are shown in Fig. 8.

TABLE III
LAND COVER CLASSES AND THE NUMBERS OF SAMPLES IN THE HOUSTON DATASET

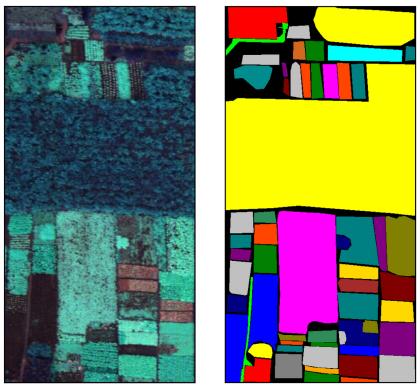
Class No.	Color	Class Name	Pixels
1		Healthy grass	1251
2		Stressed grass	1254
3		Synthetic grass	697
4		Tress	1244
5		Soil	1242
6		Water	325
7		Residential	1268
8		Commercial	1244
9		Road	1252
10		Highway	1227
11		Railway	1235
12		Parking Lot 1	1233
13		Parking Lot 2	469
14		Tennis Court	428
15		Running Track	660
		TOTAL	15029



(a)

(b)

Fig. 8. Houston dataset. (a) False color image. (b) Ground-truth map.



(a) (b)

Fig. 9. WHU-Hi-HongHu dataset. (a) False color image. (b) Ground-truth map.

4) WHU-Hi-HongHu: The WHU-Hi-HongHu dataset was collected on November 20, 2017, in Honghu, China, using a Headwall Nano-Hyperspec sensor mounted on a DJI Matrice 600 Pro UAV. The imagery, captured at 100 m altitude, has 940 × 475 pixels, 270 spectral bands (400–1000 nm), and 0.043 m spatial resolution. This dataset is a complex agricultural scene, with 386 693 pixels representing 22 land cover categories, featuring a large data volume and rich variety of classes. The details are shown in Table IV and Fig. 9.

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TABLE IV
LAND COVER CLASSES AND THE NUMBERS OF SAMPLES IN THE WHU-HI-HONGHU DATASET

Class No.	Color	Class Name	Pixels
1		Red roof	14041
2		Road	3512
3		Bare soil	21821
4		Cotton	163285
5		Cotton firewood	6218
6		Rape	44557
7		Chinese cabbage	24103
8		Pakchoi	4054
9		Cabbage	10819
10		Tuber mustard	12394
11		Brassica parachinensis	11015
12		Brassica chinensis	8954
13		Small Brassica chinensis	22507
14		Lactuca sativa	7356
15		Celtuce	1002
16		Film covered lettuce	7262
17		Romaine lettuce	3010
18		Carrot	3217
19		White radish	8712
20		Garlic sprout	3486
21		Broad bean	1328
22		Tree	4040
		TOTAL	386693

B. Parameter Settings

To guarantee the quality of data generation and the convergence of the network, we have established a training process that includes 2000 training epochs. Each training epoch consisted of five AG stages, followed by one DER stage and one RC stage respectively. We set the learning rate of the discriminator $\boldsymbol{D}$ to 0.0002, and the learning rate for the encoding module $\boldsymbol{E}$ and the generator $\boldsymbol{G}$ to 0.0001. Considering that the number of training cycles in the RC stage is fewer than the generative stage, we set the learning rate for the RC stage to 0.001 to expedite the learning process of the FSHyperRGAN. The RMSprop optimizer was employed to optimize model parameters. Within each training epoch, we sampled a N-way, K-shot classification task $\mathbf{T_i}$ from the training set. Here, N was set to the number of classes in the dataset. The number of samples for each class K in the support set is set to 1. During each training epoch, K samples from each class in the training set are sampled and placed into the support set, enabling query samples to be compared with a single support example from each class. All our experiments employed the PyTorch framework and were repeated ten times on various initial training samples.

In our experiments, two hyperparameters, namely the spatial size and the number of principal components retained during PCA dimensionality reduction, have an impact on the classification performance of 3D-FSHyperRGAN. We analyze these two hyperparameters by comparing the classification performance across four different datasets under various parameter settings.

1) *Spatial Size:* In our proposed 3D-FSHyperRGAN, the spatial size has an impact on the extraction of spatial information. In the task of HSI classification, large spatial size may lead to difficulties in feature extraction and an increased potential for feature interference. Conversely, a small spatial size may result in insufficient information extraction. Therefore, to determine the suitable spatial size for our proposed method, we calculated the OA, AA, and Kappa coefficient using different spatial sizes. From Fig. 10, it can be observed that, except for the PaviaU dataset, the classification performance of the model initially increases and then decreases with the increase in spatial size. Based on this observation, we set the spatial size to 27 $\times$ 27 in our experiments to achieve better network model performance.

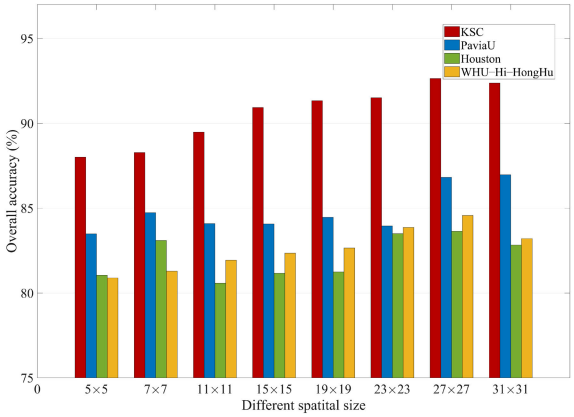


Fig. 10. Classification results of the proposed 3D-FSHyperRGAN with different spatial size.

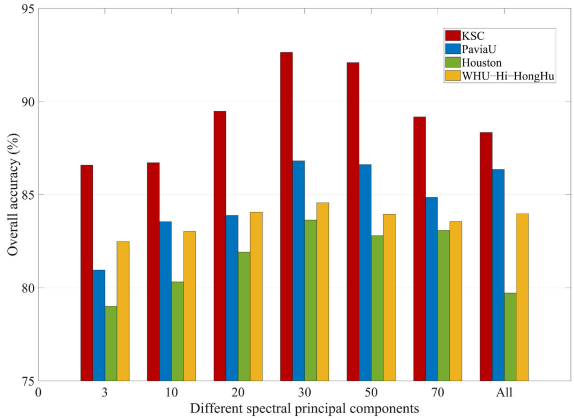


Fig. 11. Classification results of the proposed 3D-FSHyperRGAN with different spectral principal components.

2) *Spectral Principal Components:* In the spectral–spatial classifier, we PCA to reduce the dimensionality of the spectral data, ensuring the retention of sufficient spectral information. To determine the optimal number of spectral principal components for our proposed 3D-FSHyperRGAN, we conducted comparative experiments, analyzing the performance with different numbers of spectral principal components. The results are shown in Fig. 11. As can be observed from the figure, when fewer spectral principal components are used, the extracted spectral information is less comprehensive, leading to inferior model training and a decline in classification performance. However, unlike spectral classifiers that only utilize spectral information, selecting

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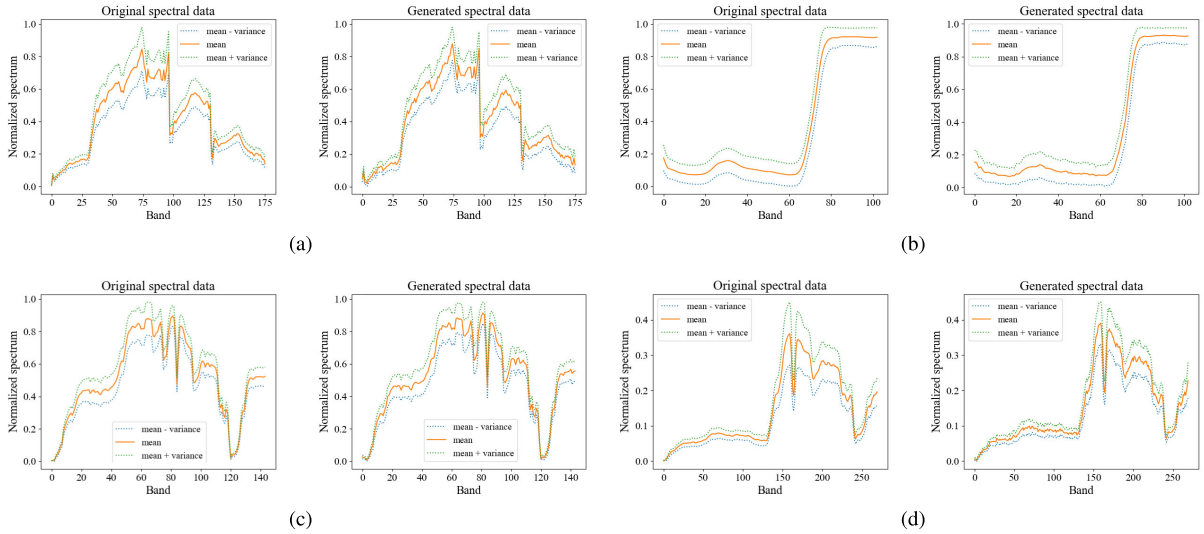


Fig. 12. Illustration of the spectral features on different datasets. (a) Spartine marsh class from KSC dataset. (b) Trees class from PaviaU dataset. (c) Running track class from Houston dataset. (d) Celtuce class from WHU-Hi-HongHu dataset.

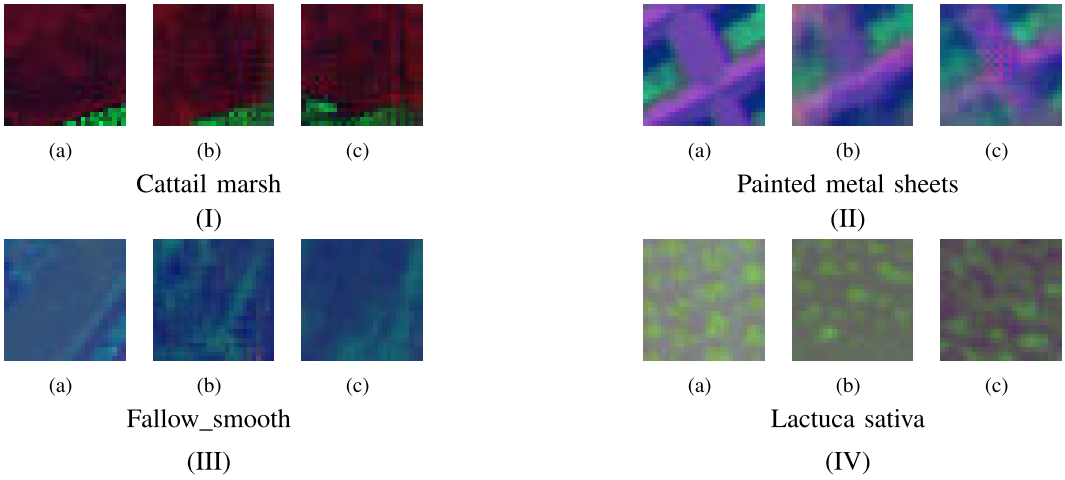


Fig. 13. False color illustration of the spatial–spectral features on different datasets. (I) KSC dataset. (II) PaviaU dataset. (III) Houston dataset. (IV) WHU-Hi-HongHu dataset. (a) Real data, (b) reconstructed data, and (c) generated data.

an excessive number of spectral principal components can cause overfitting issues when training samples are limited, thereby affecting classification performance. Moreover, as the number of spectral principal components increases, the time required for model training also escalates. Based on the experimental results, we set the number of spectral principal components to 30 in our experiments. This allows us to achieve favorable classification results while minimizing the time consumption of model training.

C. Visualization of Generated Samples

To illustrate the effectiveness of the generated spectral spectrum, we plot the generated virtual spectral spectrum in Fig. 12. It can be seen that the spectral feature distribution of the generated virtual samples is consistent with the distribution of real samples. Moreover, to demonstrate the effectiveness of the generated spatial features, we draw the corresponding virtual spatial features in Fig. 13. It can be seen that the samples generated by our method are closer to the real sample distribution. Therefore, our proposed FSHyperRGAN successfully reconstructed samples from real data, which can provide diverse data for the training process and improve the classification performance of the model in return.

D. Classification Results of 1D-FSHyperRGAN

In order to demonstrate the effectiveness of our proposed 1D-FSHyperRGAN, a series of comparative methods are compared, including random forest (RF) [56], multilayer perceptron (MLP) [57], SVM [15], 1-D convolutional neural network (1D-CNN) [58], RelatinNetWork [55], 1D-GAN [35], and 1-D convolutional capsule network (1D-Conv-Capsule) [21].

As can be observed from the results in Table V, our proposed 1D-FSHyperRGAN exhibits superior performance in terms of overall OA, AA, and Kappa coefficient across the four datasets. On the KSC and Houston datasets, the 1D-GAN and the 1D-Conv-Capsule networks, with very good feature extraction capabilities, demonstrate superior

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TABLE V

CLASSIFICATION RESULTS WITH SPECTRAL FEATURES ON DIFFERENT DATASETS (FIVE LABELED SAMPLES PER CLASS)

Data set	Metrics	RF	SVM	MLP	1D-CNN	RelatinNetwork	1D-GAN	1D-Conv-Capsule	1D-FSHyperRGAN
KSC	OA (%)	71.82±2.18	75.69±2.03	76.48±3.57	79.06±2.83	76.45±2.74	79.70±2.07	78.93±1.72	**80.27±2.36**
	AA (%)	66.47±2.70	69.53±1.68	71.31±3.50	73.99±2.87	70.95±3.01	74.29±2.95	72.72±1.31	**74.90±2.39**
	Kappa (×100)	68.68±2.37	72.94±2.23	73.92±3.88	76.75±3.11	73.84±3.01	77.39±2.31	76.58±1.87	**78.06±2.63**
PaviaU	OA (%)	57.38±5.78	58.77±5.03	59.66±6.00	61.87±7.61	66.41±3.36	63.68±9.68	65.35±3.84	**68.21±5.77**
	AA (%)	70.77±2.23	72.88±2.33	70.53±2.83	71.82±2.56	74.18±2.41	74.67±3.50	72.74±2.31	**77.05±3.01**
	Kappa (×100)	48.58±5.51	50.16±4.98	50.60±5.85	53.04±7.66	57.90±3.56	55.62±10.42	56.75±3.97	**59.49±6.29**
Houston	OA (%)	62.01±1.77	62.42±1.71	65.37±3.59	68.67±3.07	66.24±2.48	71.68±3.01	71.95±2.66	**73.47±1.93**
	AA (%)	63.75±1.86	64.01±1.64	66.75±3.66	69.53±3.50	67.83±1.82	72.92±3.05	73.02±2.43	**74.32±1.32**
	Kappa (×100)	59.06±1.91	59.49±1.85	62.65±3.89	66.19±3.31	63.58±2.65	69.44±3.23	69.70±2.87	**71.36±2.05**
WHU-Hi-HongHu	OA (%)	50.01±6.28	53.66±4.98	54.75±5.84	57.55±6.04	62.84±3.92	60.97±7.85	64.31±3.54	**67.39±4.26**
	AA (%)	53.59±1.16	51.86±2.10	52.93±2.27	57.88±2.31	68.56±2.60	58.38±3.24	70.10±1.67	**72.35±2.47**
	Kappa (×100)	43.82±5.92	46.72±5.64	47.55±5.49	51.15±5.82	54.21±3.69	52.61±9.71	56.89±2.94	**58.85±5.08**

TABLE VI

CLASSIFICATION RESULTS WITH SPATIAL–SPECTRAL FEATURES ON THE KSC DATASET (FIVE LABELED SAMPLES PER CLASS)

class	SVM	3D-CNN	3D-GAN	SACVAEGAN	SDRN	3D-Conv-Capsule	S-DMM	DCFSL	DM-MRN	ISGBA	3D-FSHyperRGAN
1	85.93±6.24	84.79±7.52	83.15±8.65	97.58±2.22	87.73±4.39	92.45±6.01	94.73±4.72	95.1±1.70	**98.81±1.32**	98.01±1.74	98.39±1.46
2	80.59±4.38	64.27±12.39	64.48±16.63	58.11±20.05	62.17±6.04	70.30±13.36	89.61±4.07	**92.41±4.28**	78.34±5.30	82.84±1.96	83.73±16.28
3	67.51±17.85	77.69±10.01	59.91±13.21	86.52±9.85	85.06±10.35	77.91±12.90	93.12±5.97	93.6±2.76	**98.67±0.38**	95.07±5.37	87.36±13.00
4	38.92±15.54	35.18±10.10	**67.68±13.29**	18.21±17.21	38.15±9.24	66.88±5.76	54.32±17.38	65.45±12.8	48.37±11.96	47.15±1.45	55.28±16.61
5	52.72±6.19	59.93±7.21	91.49±8.75	76.07±4.25	72.66±23.45	**93.68±9.71**	77.74±6.73	81.94±7.6	77.20±3.18	81.29±9.76	78.85±18.00
6	27.11±16.25	44.58±9.81	76.75±25.20	64.35±16.11	66.55±18.27	91.39±4.88	79.50±10.55	84.75±6.07	**97.61±1.65**	84.90±13.78	90.02±8.21
7	93.88±2.92	92.50±6.57	96.36±7.41	97.48±4.50	84.31±6.18	96.80±4.35	96.45±5.41	90.91±4.21	**99.33±0.95**	99.33±0.95	99.00±3.32
8	52.72±20.78	48.63±13.93	57.49±15.45	59.47±14.94	78.78±18.72	58.53±8.31	76.74±19.34	82.82±6.21	84.08±10.39	91.06±6.19	**91.31±17.05**
9	82.46±10.15	84.66±5.21	89.13±6.61	97.33±4.60	79.34±6.13	89.62±5.09	93.31±7.39	79.38±9.84	83.14±2.83	**99.87±0.09**	92.83±7.20
10	77.72±7.25	74.65±13.64	82.23±13.51	98.87±1.79	94.31±6.3	63.07±8.66	93.55±7.37	**99.5±1.68**	83.00±8.21	80.65±19.11	98.04±2.97
11	92.26±4.12	96.00±2.32	76.26±10.47	98.19±2.84	97.12±2.39	98.68±2.23	97.69±2.91	**100.0±3.78**	99.60±0.57	98.22±1.48	98.16±2.28
12	76.70±8.11	81.09±4.18	56.57±14.44	93.75±3.06	85.8±2.47	76.69±8.99	88.21±7.94	96.38±3.85	**98.93±0.34**	92.69±7.16	93.19±4.41
13	99.92±0.05	99.99±0.04	**100.00±0.00**	100.00±0.00	100.0±0.0	99.62±0.83	100.00±0.00	100.0±0.00	99.67±0.46	100.00±0.00	99.83±0.56
OA (%)	77.55±2.03	78.30±3.50	78.83±3.20	86.20±1.90	84.24±2.13	84.52±1.67	90.21±2.02	91.70±1.81	90.96±1.14	91.80±2.63	**92.64±2.29**
AA (%)	71.46±1.84	72.61±4.00	77.16±3.32	80.22±2.25	79.79±3.84	82.89±2.16	87.59±2.20	89.40±2.08	88.21±0.65	88.55±2.26	**89.59±2.98**
Kappa (×100)	75.01±2.25	75.86±3.86	76.45±3.54	84.57±2.16	82.48±2.37	82.80±1.86	89.10±2.25	90.76±2.01	89.93±1.27	90.87±2.92	**91.81±2.55**

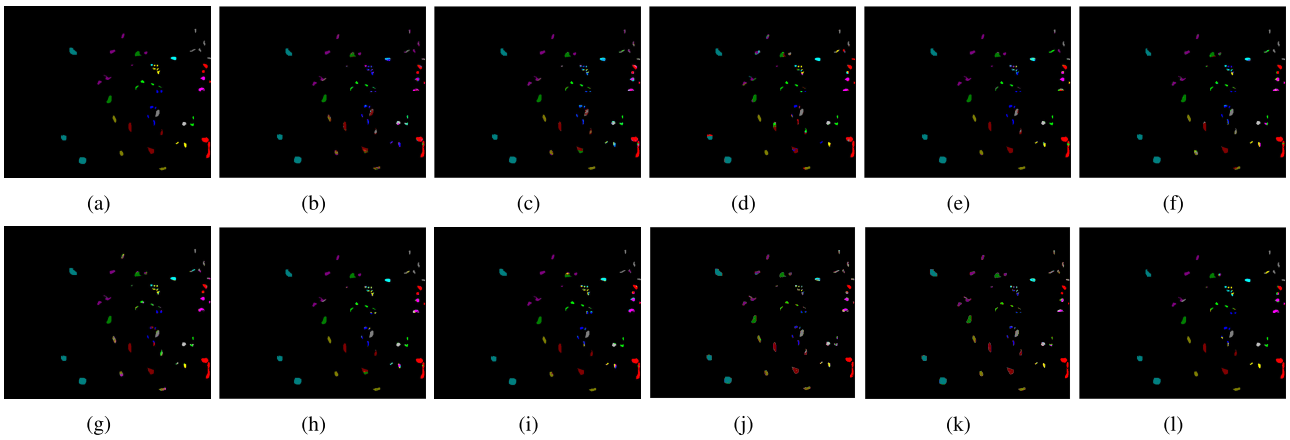


Fig. 14. Classification maps on KSC dataset. (a) Ground truth. (b) SVM. (c) 3D-CNN. (d) 3D-GAN. (e) SACVAEGAN. (f) SDRN. (g) 3D-Conv-Capsule. (h) S-DMM. (i) DCFSL. (j) DM-MRN. (k) ISGBA. (l) 3D-FSHyperRGAN.

classification results compared to the RN. Meanwhile, our proposed 1D-FSHyperRGAN gets better classification accuracy. This demonstrates that our proposed 1D-FSHyperRGAN can generate more spectral samples and guide the training process.

E. Classification Results of 3D-FSHyperRGAN

In the 3D-FSHyperRGAN, we fully exploited both the spectral and spatial information to enhance the classification results. To illustrate the effectiveness of the proposed 3D-FSHyperRGAN, we compared 3D-FSHyperRGAN with relevant methods, including SVM [6], 3D-CNN [20], 3D-GAN [35], SDRN [18], 3D-Conv-Capsule [21], SACVAEGAN [40], S-DMM [48], DCFSL [51], DM-MRN [53], and ISGBA [41]. Since the source domain data of DCFSL only has 19 categories, the DCFSL method is not compared for the WHU-Hi-HongHu dataset.

1) Results on the KSC Dataset: Table VI displays the classification results on the KSC dataset. From Table VI, we can see that the deep learning methods, such as 3D-CNN, SDRN, and 3D-Conv-Capsule, outperform traditional methods such as SVM. The FSL methods such as S-DMM, DCFSL, DM-MRN, and ISGBA achieve higher classification accuracy than the deep learning methods. Specifically, the ISGBA method improves OA by 7.56% compared to the SDRN method and 7.28% when compared to the 3D-Conv-Capsule method. Furthermore, our proposed 3D-FSHyperRGAN not only benefits from FSL but also the

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TABLE VII

CLASSIFICATION RESULTS WITH SPATIAL–SPECTRAL FEATURES ON THE PAVIAU DATASET (FIVE LABELED SAMPLES PER CLASS)

class	SVM	3D-CNN	3D-GAN	SACVAEGAN	SDRN	3D-Conv-Capsule	S-DMM	DCFSL	DM-MRN	ISGBA	3D-FSHyperRGAN
1	50.38±20.02	49.90±13.53	61.30±5.74	68.75±4.26	71.31±25.55	83.75±5.74	85.32±9.17	82.76±5.25	72.12±7.15	65.89±5.59	**85.79±10.15**
2	56.20±16.84	69.32±10.68	61.62±7.70	68.06±4.50	70.66±7.11	73.05±13.78	65.04±14.47	80.50±8.96	83.13±6.53	**86.99±1.54**	85.33±8.19
3	31.21±19.79	55.46±9.55	62.65±7.49	69.91±2.50	72.7±13.94	64.07±5.87	78.70±11.11	63.30±7.69	91.94±2.75	**93.17±0.36**	85.79±7.78
4	92.20±4.94	81.89±14.38	71.97±8.80	94.60±1.59	95.57±1.79	83.79±10.37	92.15±5.82	94.07±3.82	92.82±5.46	**96.53±4.07**	94.02±5.23
5	98.97±0.31	98.26±2.00	84.64±13.80	99.89±0.19	**100.0±0.0**	96.31±5.42	98.52±4.08	97.97±2.10	99.93±0.09	99.93±0.34	99.77±0.23
6	51.28±14.93	40.01±10.17	76.63±12.22	80.35±17.81	66.86±7.86	81.11±12.59	76.13±10.47	75.41±6.46	90.17±12.19	**98.00±2.89**	86.59±6.75
7	92.80±3.65	75.61±10.63	85.09±4.46	82.42±6.70	92.38±7.82	98.10±2.87	95.40±2.90	79.56±1.30	**98.31±1.29**	98.00±0.26	94.48±6.22
8	69.67±17.86	48.12±10.38	69.64±10.73	77.70±12.63	**90.56±8.44**	71.80±11.78	82.09±11.91	58.26±15.79	55.31±12.07	42.41±12.50	80.63±9.82
9	99.88±0.10	96.18±3.99	86.15±7.15	87.58±10.89	**99.98±0.09**	90.98±8.30	99.97±0.05	98.83±0.98	97.43±2.84	98.89±0.74	98.63±1.66
OA (%)	60.67±6.51	62.96±3.54	67.13±4.08	75.08±3.08	76.68±3.62	78.74±4.71	76.17±6.50	79.39±4.18	82.29±2.56	83.17±2.01	**86.82±3.20**
AA (%)	71.40±1.58	68.31±3.02	73.62±2.24	81.73±2.99	85.42±3.29	82.90±2.17	85.54±2.52	81.19±1.87	86.80±0.79	86.65±1.52	**90.11±1.78**
Kappa (×100)	51.64±6.06	53.03±3.65	59.60±4.46	68.91±3.66	70.65±4.30	72.94±5.30	70.36±7.13	73.62±4.88	77.39±2.92	78.36±2.50	**82.97±3.82**

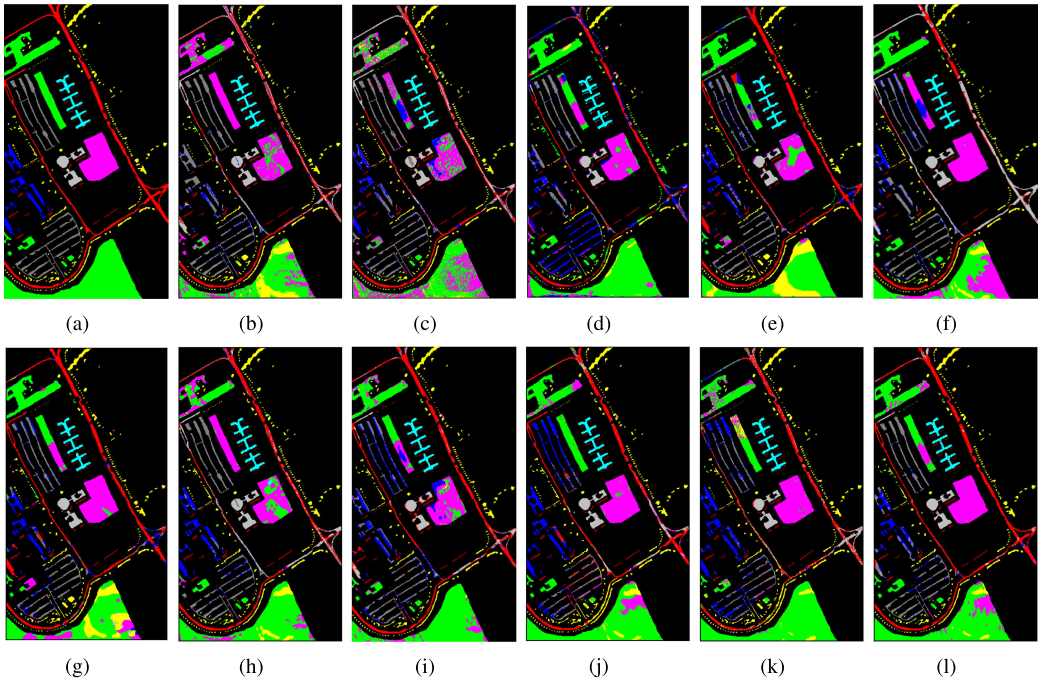


Fig. 15. Classification maps on PaviaU dataset. (a) Ground truth. (b) SVM. (c) 3D-CNN. (d) 3D-GAN. (e) SACVAEGAN. (f) SDRN. (g) 3D-Conv-Capsule. (h) S-DMM. (i) DCFSL. (j) DM-MRN. (k) ISGBA. (l) 3D-FSHyperRGAN.

augmentation of the generated data, which shows a 0.81% improvement over the well-performing ISGBA method in terms of OA. Compared to other state-of-the-art methods, the proposed 3D-FSHyperRGAN achieved the best results in OA, AA, and Kappa coefficient. The classification maps of different methods from Fig. 14 demonstrate that, compared to other methods, our proposed 3D-FSHyperRGAN gets smoother results.

2) Results on the PaviaU Dataset: Table VII presents the experimental results on the PaviaU dataset. Compared to other methods, the proposed method has made significant improvements in OA, AA, and Kappa. Our proposed 3D-FSHyperRGAN improved OA, AA, and Kappa by 3.65%, 3.46%, and 4.61%, respectively, compared to the second-best ISGBA method. This is because there are fewer categories and more testing samples in the PaviaU dataset. Our proposed 3D-FSHyperRGAN can generate more virtual samples for training and achieve better results. Fig. 15 presents the classification maps of different methods. It is clear that our proposed method generates accurate boundaries.

3) Results on the Houston Dataset: Table VIII shows the classification results of the Houston dataset. From Table VIII, it can be seen that our proposed 3D-FSHyperRGAN improved OA by 1.85%, AA by 0.86%, and Kappa by 1.99% compared to the DM-MRN method. This strongly suggests that the samples generated by our proposed method can be effectively used in RN to learn richer spectral and spatial information, thereby improving classification accuracy. Fig. 16 displays the classification maps of different methods. From Fig. 16, it can be seen that our proposed method handles classification boundaries better and generates smoother classification results.

4) Results on the WHU-Hi-HongHu Dataset: The experimental results on the WHU-Hi-HongHu dataset are listed in Table IX. From Table IX, our method has the best classification performance on most categories, such as Carrot and White radish, indicating that generated samples can benefit the RN. Additionally, compared to the second-best DM-MRN method, our proposed 3D-FSHyperRGAN improved OA by 1.98%, AA by 4.94%, and Kappa by 2.49%, demonstrating superior performance for few-shot classification. The classification maps of different methods in Fig. 17 show that the map of our proposed method is closer to the ground truth and has more accurate classification for edge points.

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TABLE VIII

CLASSIFICATION RESULTS WITH SPATIAL–SPECTRAL FEATURES ON THE HOUSTON DATASET (FIVE LABELED SAMPLES PER CLASS)

class	SVM	3D-CNN	3D-GAN	SACVAEGAN	SDRN	3D-Conv-Capsule	S-DMM	DCFSL	DM-MRN	ISGBA	3D-FSHyperRGAN
1	85.33±1.64	82.93±10.75	73.23±5.87	80.04±2.31	87.64±4.47	85.89±3.98	90.73±6.01	89.47±4.76	91.09±7.48	93.62±5.94	**93.86±4.58**
2	66.89±7.68	58.76±13.28	49.15±6.01	58.81±2.74	**97.87±2.42**	81.64±11.39	88.31±9.79	84.20±8.59	85.09±5.04	85.16±15.05	89.36±8.72
3	98.03±0.71	88.57±8.99	91.71±5.83	97.93±0.71	99.55±0.46	99.10±1.67	**99.66±0.48**	98.06±0.86	97.92±1.23	99.46±0.41	98.45±1.43
4	85.38±9.15	73.39±21.30	48.41±8.56	81.78±15.13	97.1±3.84	76.43±14.29	**97.48±3.17**	94.21±3.97	89.56±6.69	94.44±5.27	96.90±2.65
5	74.20±10.53	75.05±11.59	76.19±7.55	86.45±3.46	95.44±6.96	86.25±7.24	**98.49±1.69**	93.26±2.75	96.78±6.31	94.72±7.53	95.54±7.71
6	72.64±0.69	65.40±12.88	84.78±7.29	76.09±2.77	84.91±3.07	82.91±5.86	**91.22±6.60**	82.44±7.04	89.88±6.92	83.35±8.82	86.65±8.35
7	30.86±22.90	45.64±15.40	63.67±11.23	82.05±5.71	76.29±7.74	69.93±8.60	71.51±13.93	77.69±9.24	**85.18±4.84**	76.19±7.14	83.96±11.66
8	25.94±14.84	45.34±12.90	42.82±10.77	41.17±3.00	41.89±12.58	**59.34±9.58**	54.66±12.92	49.86±4.97	47.96±9.86	51.11±9.11	55.46±11.20
9	52.34±26.18	43.09±18.29	47.63±5.71	**79.34±1.14**	72.6±3.62	60.91±8.56	70.81±17.62	66.83±13.68	71.08±7.12	76.63±12.55	73.76±7.02
10	5.49±9.52	46.82±14.91	**83.63±6.55**	47.60±8.51	69.89±16.53	74.17±6.95	68.94±8.70	64.34±10.26	75.16±12.99	77.74±15.43	78.17±16.69
11	66.20±10.28	39.23±13.42	71.12±10.63	49.06±14.41	73.38±16.39	61.07±12.15	72.26±17.81	64.70±6.61	77.89±10.42	71.80±10.68	**78.35±14.08**
12	14.28±10.89	36.57±10.93	60.81±6.16	38.18±7.55	42.11±9.4	71.67±11.50	44.35±13.01	**74.06±7.48**	69.09±12.88	54.31±13.42	72.31±10.65
13	3.89±5.00	29.96±6.64	72.13±11.54	82.20±4.62	50.14±12.86	77.07±7.10	63.87±21.44	**88.02±5.91**	87.33±5.70	80.68±8.51	78.55±13.79
14	88.95±8.67	73.67±16.93	97.68±3.65	94.91±4.11	94.47±10.63	87.79±9.89	98.50±4.17	91.54±4.99	**99.95±0.09**	98.88±1.60	98.56±2.48
15	91.99±2.99	80.83±17.67	76.16±10.49	98.79±1.43	99.65±0.33	74.53±17.46	**99.71±0.22**	96.89±3.06	99.57±0.85	99.29±1.20	99.24±0.92
OA (%)	54.91±1.91	56.95±5.96	65.06±3.04	69.13±3.91	77.46±3.06	75.03±3.98	78.49±2.64	78.80±2.90	81.79±2.56	80.32±3.18	**83.64±2.65**
AA (%)	57.55±1.90	58.45±6.07	68.91±2.84	72.88±3.60	78.72±2.35	76.88±3.61	80.53±2.53	81.04±2.58	84.23±2.30	82.48±2.67	**85.09±1.90**
Kappa (×100)	51.39±2.06	53.48±6.45	62.32±3.26	66.74±4.17	75.64±3.29	73.04±4.30	76.77±2.85	77.10±3.12	80.33±2.77	78.76±3.42	**82.32±2.85**

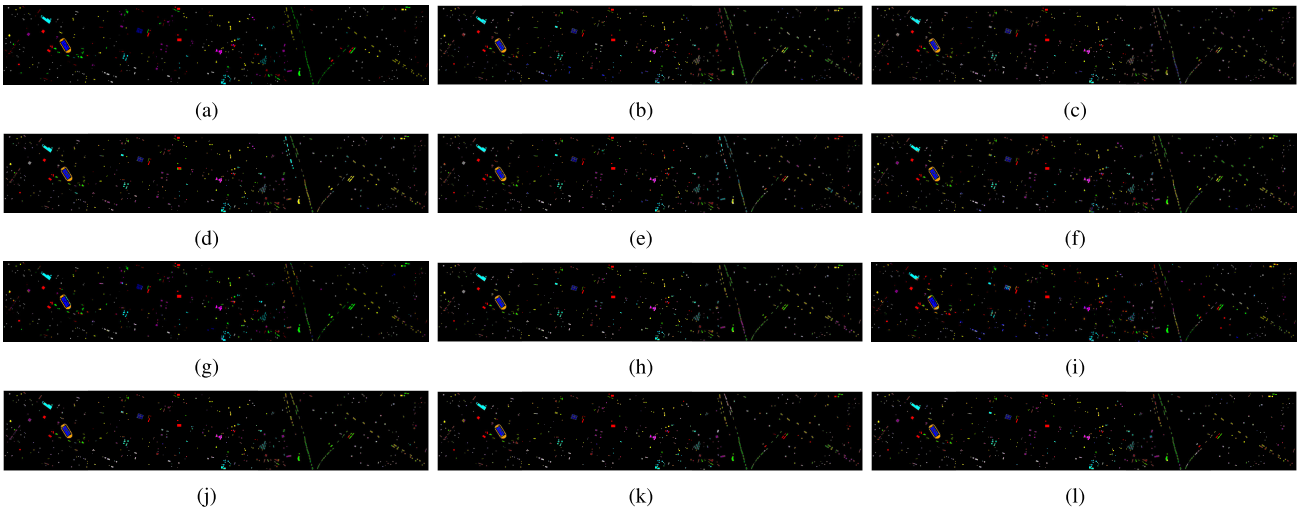


Fig. 16. Classification maps on Houston dataset. (a) Ground truth. (b) SVM. (c) 3D-CNN. (d) 3D-GAN. (e) SACVAEGAN. (f) SDRN. (g) 3D-Conv-Capsule. (h) S-DMM. (i) DCFSL. (j) DM-MRN. (k) ISGBA. (l) 3D-FSHyperRGAN.

TABLE IX

CLASSIFICATION RESULTS WITH SPATIAL–SPECTRAL FEATURES ON THE WHU-HI-HONGHU DATASET (FIVE LABELED SAMPLES PER CLASS)

class	SVM	3D-CNN	3D-GAN	SACVAEGAN	SDRN	3D-Conv-Capsule	S-DMM	DM-MRN	ISGBA	3D-FSHyperRGAN
1	72.86±6.99	88.97±18.02	91.85±15.10	81.85±3.23	**96.13±8.91**	64.21±14.20	75.28±4.99	75.59±10.98	88.90±6.74	89.76±3.60
2	65.18±4.72	73.74±13.02	80.81±2.35	71.53±4.04	86.19±10.45	60.73±14.82	**88.12±5.25**	69.70±3.05	68.43±22.32	86.17±3.21
3	71.40±3.24	74.17±3.21	76.45±8.51	80.18±5.23	75.08±7.87	**83.36±3.49**	82.55±11.20	82.28±10.25	83.33±3.22	83.03±3.45
4	66.65±10.82	69.45±15.16	82.20±8.29	82.15±7.28	83.37±11.79	84.05±4.13	88.53±3.91	89.62±7.09	84.58±4.93	**89.91±5.23**
5	57.80±9.92	58.60±7.65	69.39±11.16	83.75±4.77	72.38±9.23	73.87±5.39	84.59±2.19	77.11±9.35	91.49±3.73	**92.54±6.86**
6	76.71±4.05	80.74±1.66	84.21±8.83	90.77±1.06	86.08±2.58	92.04±0.27	91.31±6.24	**96.11±5.06**	91.11±1.19	93.12±2.31
7	40.32±12.90	56.15±12.42	38.88±5.99	51.80±16.40	45.82±11.57	49.71±14.20	31.29±11.40	63.38±2.82	59.84±6.75	**64.02±4.36**
8	17.72±8.12	16.84±3.43	11.30±5.38	39.20±7.40	3.157±9.95	27.43±2.98	35.09±7.74	47.68±2.47	**56.15±6.60**	55.39±5.34
9	84.00±5.55	90.35±3.29	94.96±8.28	94.71±2.37	**97.54±3.95**	93.88±0.19	92.77±3.31	94.39±4.13	94.93±1.87	94.05±1.48
10	31.67±4.75	32.86±7.46	21.66±8.01	59.07±15.70	10.74±9.63	36.75±8.86	60.33±8.88	**68.83±7.89**	61.13±7.79	63.35±9.42
11	21.76±9.93	44.51±18.07	22.16±8.54	33.32±10.83	12.62±11.16	33.85±18.34	46.63±10.20	33.71±2.70	45.53±14.28	**47.19±8.58**
12	51.07±3.70	58.58±15.91	59.46±20.69	55.05±9.15	65.12±4.31	43.74±19.61	59.53±10.40	48.53±3.87	55.58±10.79	**69.98±1.32**
13	37.72±8.20	53.49±17.46	54.24±10.91	58.65±11.45	55.55±5.58	64.96±3.73	60.11±6.74	69.26±7.82	67.30±5.21	**69.32±11.1**
14	47.91±5.28	76.49±8.12	67.86±18.14	74.45±9.41	81.66±7.00	72.71±6.33	**89.28±3.17**	82.35±2.10	79.80±10.67	87.24±3.22
15	68.81±4.34	82.43±1.20	89.27±9.42	97.86±2.49	94.51±2.00	97.24±1.28	**98.86±5.79**	97.21±7.23	96.13±4.75	98.84±1.86
16	58.07±9.18	66.62±7.21	85.28±6.86	85.07±6.92	87.04±3.67	86.33±6.85	90.12±7.47	87.57±7.05	92.06±2.88	**92.89±8.26**
17	66.63±3.23	76.87±18.40	80.15±4.72	93.23±2.77	82.09±4.91	82.20±11.83	88.87±10.10	**99.34±5.93**	89.66±10.36	96.64±5.07
18	59.04±4.37	85.17±10.92	77.78±9.38	79.71±11.86	76.18±8.81	70.23±3.77	83.25±5.32	81.19±13.81	90.79±3.09	**93.05±6.36**
19	50.11±11.31	90.04±20.66	83.72±10.51	82.50±5.07	83.13±7.41	84.16±5.90	89.70±7.30	84.83±12.79	89.67±4.17	**92.76±3.96**
20	65.47±6.81	83.99±16.57	70.52±6.84	77.75±9.68	78.16±13.63	84.00±6.90	91.39±8.22	94.87±13.76	93.91±2.14	**95.65±5.17**
21	66.16±7.52	48.49±14.64	70.59±15.68	72.73±5.60	63.32±15.39	67.92±18.05	75.49±12.24	86.67±3.54	81.14±14.46	**86.90±5.04**
22	61.93±8.98	68.29±9.68	82.47±9.50	95.71±7.85	81.06±1.94	80.10±12.38	94.09±5.60	**97.85±7.09**	96.66±1.51	94.90±2.88
OA (%)	60.04±5.38	68.55±7.47	72.75±4.75	76.93±2.25	74.21±5.74	76.03±2.37	79.63±1.93	82.59±3.27	80.77±2.95	**84.57±2.31**
AA (%)	56.47±0.78	67.13±3.46	67.77±1.69	74.59±2.00	68.96±1.78	69.70±2.47	77.14±1.34	78.55±2.37	79.91±1.44	**83.49±1.15**
Kappa (×100)	53.23±5.09	62.83±7.66	66.96±4.21	72.11±2.34	68.64±6.25	70.72±2.67	74.90±1.85	78.47±2.94	76.56±3.38	**80.96±2.26**

To further validate the effectiveness of our proposed method, we conducted the experiments across varying numbers of labeled samples. We conducted experiments with different numbers of labeled samples per class ranging from 3 to 11. We used OA as the evaluation metric to assess the performance of different methods, and the experimental results are presented in Fig. 18. The results of the experiments demonstrate that as the number of training samples increases, all methods show improvements in terms of classification performance. However, our method exhibits superior classification performance across different numbers of labeled samples, demonstrating its

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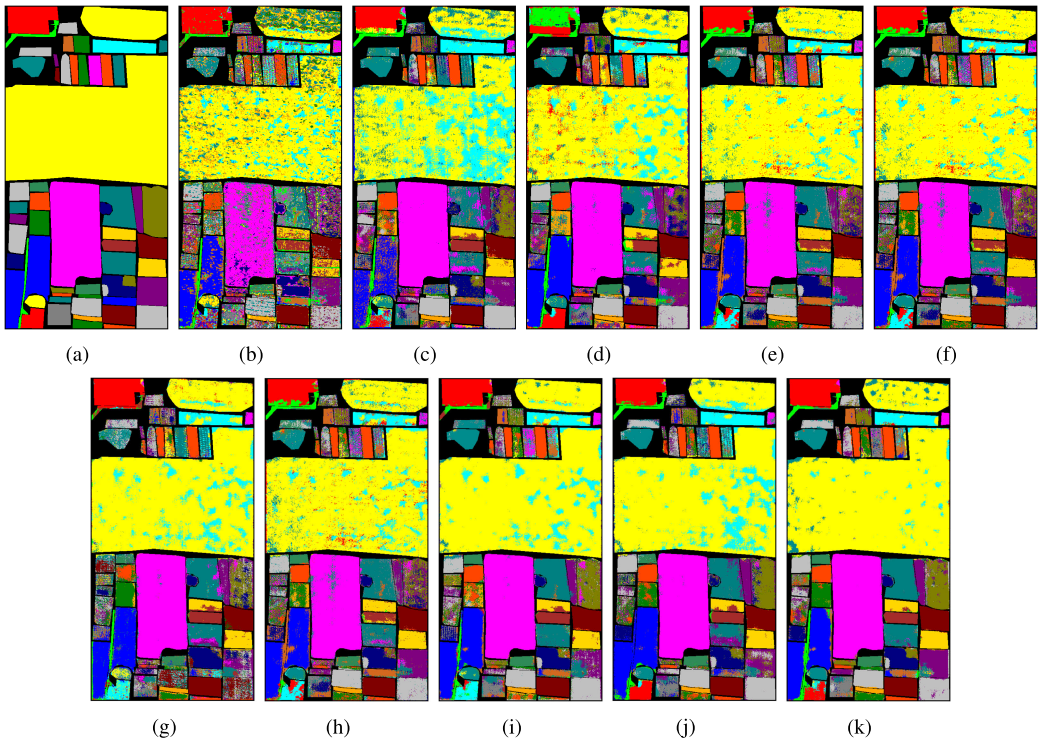


Fig. 17. Classification maps on WHU-Hi-HongHu dataset. (a) Ground truth. (b) SVM. (c) 3D-CNN. (d) 3D-GAN. (e) SACVAEGAN. (f) SDRN. (g) 3D-Conv-Capsule. (h) S-DMM. (i) DM-MRN. (j) ISGBA. (k) 3D-FSHyperRGAN.

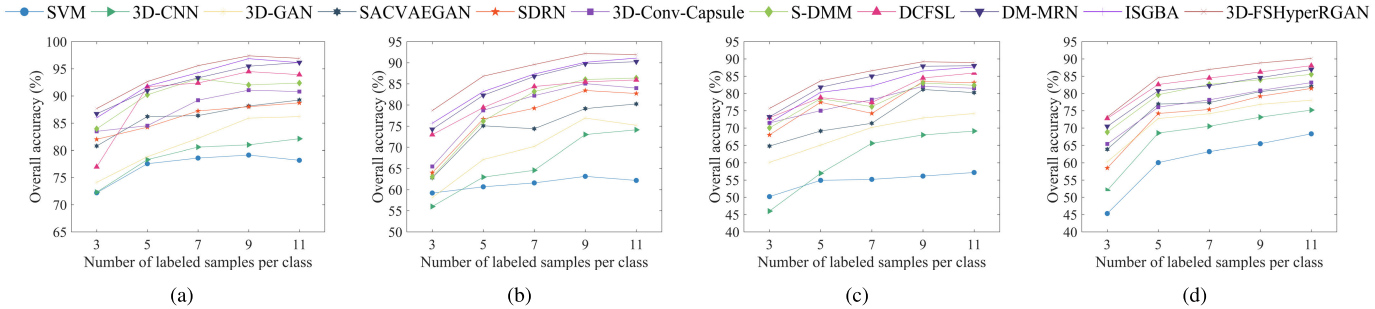


Fig. 18. Classification accuracies with different number of labeled samples on four datasets. (a) KSC. (b) PaviaU. (c) Houston. (d) WHU-Hi-HongHu.

robustness and stability across various numbers of labeled samples.

F. Ablation Study

To further validate the impact of each modules in our proposed FSHyperRGAN, we conducted ablation studies under identical experimental conditions. Using 3D-FSHyperRGAN as an example, we designed five configurations to evaluate the corresponding performances by selectively removing the AG, DER, and RC modules. These configurations include RC, DER + RC, AG + RC, and DER + AG + Classifier, where DER + AG + Classifier replaces the RC module with a multilayer convolutional neural network as the classifier.

Table X presents the experimental results for the KSC, Houston, PaviaU, and WHU-Hi-HongHu datasets, utilizing five training samples per class. The results demonstrate that RC significantly outperforms conventional classifiers in terms of OA, AA, and Kappa metrics, underscoring its crucial role in classification performance under limited sample conditions. In the RC ablation experiment, the RC module's performance closely resembles that of the S-DMM [48] method, which employs a similar relational network approach.

As observed in Table X, both AG + RC and DER + RC configurations exhibit substantial improvements in accuracy compared to the RC module alone. This finding confirms that the samples generated by AG and DER modules positively influence classification accuracy, effectively achieving data augmentation. The DER + AG + RC configuration further enhances classification performance, indicating that the integration of DER and AG modules enables the synthesis of higher quality HSI samples under RC module guidance, even with limited labeled training samples.

Based on the above analysis, we conclude that the AG, DER, and RC modules each contribute to varying degrees of performance improvement, collectively enhancing the effectiveness of our proposed FSHyperRGAN.

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TABLE X
ABLATION STUDY OF DIFFERENT MODULES IN 3D-FSHYPERRGAN

Data sets	RC	DER+RC	AG+RC	DER+AG+Classfier	DER+AG+RC
KSC	90.30±2.28	91.88±2.16	91.15±2.42	89.94±3.26	**92.64±2.29**
Houston	78.58±2.46	81.49±2.83	80.08±3.28	78.75±3.15	**83.64±2.65**
PaviaU	77.17±5.50	85.69±4.14	85.43±5.17	84.32±4.20	**86.82±3.20**
WHU-Hi-HongHu	79.86±1.83	81.84±2.65	81.15±3.14	79.27±2.87	**84.57±2.31**

TABLE XI
RUNNING TIME AND PARAMETER SIZE ON EACH DATASET WITH DIFFERENT METHODS

Data Sets	Metrics	SVM	3D-CNN	3D-GAN	SACVAEGAN	SDRN	3D-Conv-Capsule	S-DMM	DCFSL	DM-MRN	ISGBA	3D-FSHyperRGAN
Salinas	Train Time (s)	0.56	150.60	324.46	726.20	73.62	230.82	400.49	4960.05	86.46	528.41	472.58
	Test Time (s)	0.93	25.34	80.27	96.67	3.54	40.57	20.56	62.36	70.75	60.63	26.92
	Parameters (K)	-	642.72	451.35	1265.48	514.67	854.30	42.38	4056.26	190.06	1828.45	484.50
Houston	Train Time (s)	0.46	130.47	315.24	634.93	68.38	218.21	327.52	1986.70	79.40	504.18	403.50
	Test Time (s)	0.67	12.54	35.48	43.60	1.52	10.64	3.69	21.30	18.34	13.81	7.94
	Parameters (K)	-	128.12	430.67	1224.92	489.61	817.02	41.12	4038.89	186.14	1768.47	472.28
PaviaU	Train Time (s)	0.42	126.38	297.65	561.35	60.95	189.17	298.68	2269.27	75.46	417.40	386.26
	Test Time (s)	0.68	22.81	68.94	74.65	2.74	31.82	15.10	56.07	61.41	49.61	22.68
	Parameters (K)	-	126.43	427.51	1215.03	478.66	782.19	40.75	4018.61	182.78	1735.80	469.72
WHU-Hi-HongHu	Train Time (s)	0.75	190.94	389.49	807.65	89.47	317.48	526.15	-	98.47	656.76	576.14
	Test Time (s)	1.62	105.43	615.30	712.15	26.86	305.42	148.60	-	483.26	416.50	232.08
	Parameters (K)	-	650.18	472.20	1301.36	528.76	863.19	44.64	-	196.45	1986.48	501.46

G. Complexity Analysis

To analyze the computational complexity of FSHyperRGAN, we used 3D-FSHyperRGAN as a case study and compared its runtime and parameter size with other methods on four datasets, using five labeled samples per class. The generation of training samples is a time-consuming process, and GAN-based methods typically require more time to converge. Consequently, methods employing sample generation generally have longer training times compared to other deep learning or FSL approaches. As shown in Table XI, our method demonstrates good training times relative to GAN-based methods such as 3D-GAN, SACVAEGAN, and ISGBA. Furthermore, during the testing phase, we directly utilize the RC module for classification, achieving superior execution time compared to other nongenerative methods. Table XI also illustrates that the proposed 3D-FSHyperRGAN ranks moderately in terms of parameter count. Overall, our FSHyperRGAN method exhibits acceptable training times and parameter quantities, while offering rapid testing speeds and superior classification performance.

V. CONCLUSION

In this article, we have proposed a novel method FSHyperRGAN for few-shot HIC. In contrast to current methods that utilize the generative model or FSL methods, separately, FSHyperRGAN integrates the generative model and the relational network together. Furthermore, a relational feature matching scheme is also designed to use virtual samples to guide classification. Additionally, FSHyperRGAN integrates a class feature extractor and relation calculator to enhance classification accuracy. Experimental results on four datasets demonstrate that FSHyperRGAN has a significant improvement in the performance for few-shot HSI classification. For our proposed FSHyperGAN, we only used generated samples to solve the problem of limited labeled data. In future work, we will further explore cross-domain scenarios to help solve the problem of limited labeled samples. Moreover, multiple feature merging methods in RN will also be investigated in the future.

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